

CDC BRFSS 2024

457,670 Respondents

301 Variables

Public Health Survey

~30%

of deaths in Kenya are
due to CVD (WHO)

ROC-AUC: 0.8478

Best Model Performance

3 ML Models Compared

Logistic Regression · Random Forest · Gradient Boosting

CARDIOVASCULAR DISEASE RISK PREDICTION

*A Machine Learning Approach to
Cardiovascular Prevention for Kenya*

Proof-of-Concept Using CDC BRFSS 2024 Data

Capstone Project — Data Science Program

Presentation Agenda

What we will cover today

01

Problem Statement

CVD burden in Kenya & the screening gap

02

Dataset

CDC BRFSS 2024 — why it works as a proxy

03

Data Preparation

Feature engineering & BRFSS variable cleaning

04

Exploratory Analysis

Key patterns linking risk factors to CVD

05

Machine Learning

3 models, iterative improvement, tuning

06

Results & Metrics

ROC-AUC 0.85, top risk drivers

07

Recommendations

What this means for Kenya stakeholders

08

Next Steps

Roadmap to Kenyan deployment

01 THE PROBLEM

Cardiovascular Disease: Kenya's Growing Crisis

Why this problem matters — and why it's urgent

~30%

of all deaths in Kenya
attributed to CVD (WHO)

47

county health departments
need data-driven tools

No lab needed

Self-reported tool works
at community level

The Screening Gap

Most Kenyans interact with the health system **only when acutely ill**. Preventive CVD screening is rare at community level.

No simple, deployable screening tool exists that works without lab tests.

Key Drivers of Rising CVD in Kenya

- Rapid urbanization → processed, high-sodium diets
- Tobacco use prevalence (~20% of adults smoke)
- Physical inactivity increasing in urban settings
- Rising obesity (30–40% urban overweight/obese)
- Under-diagnosed diabetes & hypertension

Who Benefits? Stakeholders Across Kenya's Health System



Ministry of Health

NCD Division

Feature importance guides national NCD budget allocation



Healthcare Providers

Clinics & Hospitals

Opportunistic screening during routine visits — no labs



NHIF / SHA / Insurers

Health Financing

Risk-tier segmentation for preventive wellness benefits



County Health Depts

All 47 Counties

Risk calculator supports county-level screening drives



CHVs

Community Health Volunteers

Household-level triage tool for high-risk referral



NCD Researchers

Kenyan Universities

Replicable pipeline for Kenyan STEPwise & KDHS data

CDC BRFSS 2024 — The World's Largest Health Survey

Why this dataset is the right proxy for Kenya

457,670

Survey respondents
(2024 data)

301

Variables including
behavioral & clinical

XPT

SAS Transport format
Proper column names

Public

Free CDC download
Fully documented

Why BRFSS Works for Kenya

- ✓ **Same risk factors** Smoking, BMI, inactivity, diabetes — identical to Kenya's CVD drivers
- ✓ **Self-report method** Matches what CHVs can collect at household level in Kenya
- ✓ **Large sample** Sufficient for ML with class imbalance (~5.8% CVD positive)
- ✓ **Clear outcome** CVDINFR4: 'Ever told you had a heart attack?' (direct match)
- ✓ **Open pipeline** Entire methodology reusable with Kenya STEPwise data

⚠ Important Disclaimer

This project is PROOF-OF-CONCEPT only.
BRFSS surveys U.S. adults — not Kenyans.

Key differences requiring future recalibration:

- U.S. CVD prevalence ~8% vs Kenya ~3-5%
- Different obesity, smoking & diabetes rates
- Different healthcare access and diagnostics

→ Retrain on Kenya STEPwise data before
any operational deployment.

Feature Engineering & Data Cleaning

How raw BRFSS survey data is transformed into model-ready features

Target Variable:

has_heart_disease = 1 → CVDINFR4 == 1 ("Ever told you had a heart attack?")
22,672 positive (5.83%) | 431,067 negative (94.17%)

Demographic

_AGEG5YR — Age group (18–80+)
_SEX — Biological sex

Behavioral / Lifestyle

SMOKE100 — Ever smoked 100+ cigs
SMOKDAY2 — Current smoking freq
EXERANY2 — Any exercise past 30d
ALCDAY4 — Alcohol days per week
_TOTINDA — Physical activity level

Clinical Conditions

DIABETE4 — Diabetes diagnosis
GENHLTH — Self-rated health 1–5
PHYSHLTH — Days physical health bad
MENTHLTH — Days mental health bad

Body & Access

_BMI5 — Calculated BMI
WEIGHT2 — Body weight
HEIGHT3 — Body height
CHECKUP1 — Last routine checkup
MEDCOST1 — Couldn't afford MD

1

Mode imputation for missing values

2

Remove rows >50% missing

3

Stratified 80/20 train-test split

4

StandardScaler for LR features

5

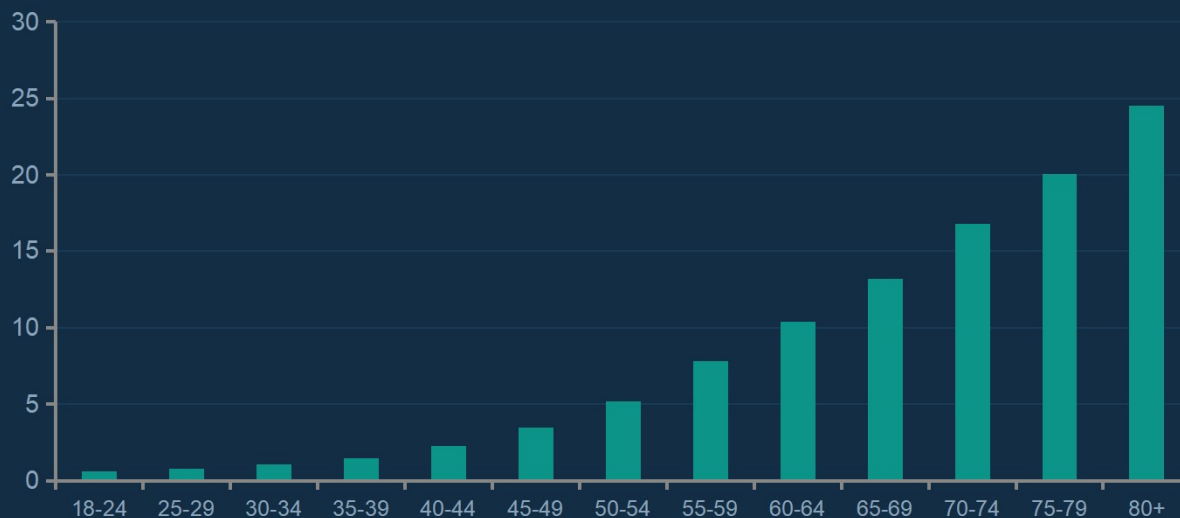
SMOTE class balancing

04 EXPLORATORY ANALYSIS

Key Patterns: Risk Factors & Heart Disease

What the data reveals before modeling

CVD Rate (%) by Age Group



$r = 0.27$ Age $\leftrightarrow$ CVD

Strongest single predictor. Risk grows 40x from 18-24 to 80+.

$r = 0.16$ GEN_HEALTH $\leftrightarrow$ CVD

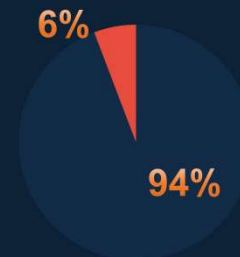
Self-rated poor health predicts disease independently.

2.1× Diabetics vs Non

Diabetics have 2.1x higher CVD rate — critical comorbidity.

Class Imbalance Challenge

Only 5.83% of respondents reported a heart attack (22,672 of 457,670). Standard classifiers trained naively will predict 'No CVD' for everyone, achieving 94% accuracy with zero clinical utility. We address this with SMOTE oversampling + class_weight='balanced'.



Iterative Model Building Strategy

From baseline to best model — each iteration justified by prior results

01

Dummy Classifier

Baseline

Always predict majority class
(No Disease)

ROC-AUC

0.500

Why: Establishes random-chance floor

02

Logistic Regression

Linear Model

SMOTE + StandardScaler
Balanced class weight

ROC-AUC

0.8066

Why: Linear baseline with interpretable coefficients

03

Random Forest

Ensemble Trees

100 trees, max_depth=15
class_weight='balanced'

ROC-AUC

0.8478

Why: Non-linear; captures interactions; more robust than single tree

04

Gradient Boosting

Boosted Ensemble

100 trees, depth=5
learning_rate=0.1

ROC-AUC

0.8061

Why: Sequential correction of errors; compared vs RF to confirm winner

★ Selected Final Model: Random Forest (highest ROC-AUC = 0.8478) → Tuned with GridSearchCV

06 RESULTS

Model Performance Results

Tuned Random Forest evaluated on holdout test set

0.8478

ROC-AUC
Best model (tuned RF)

79.80%

Accuracy
Overall correct predictions

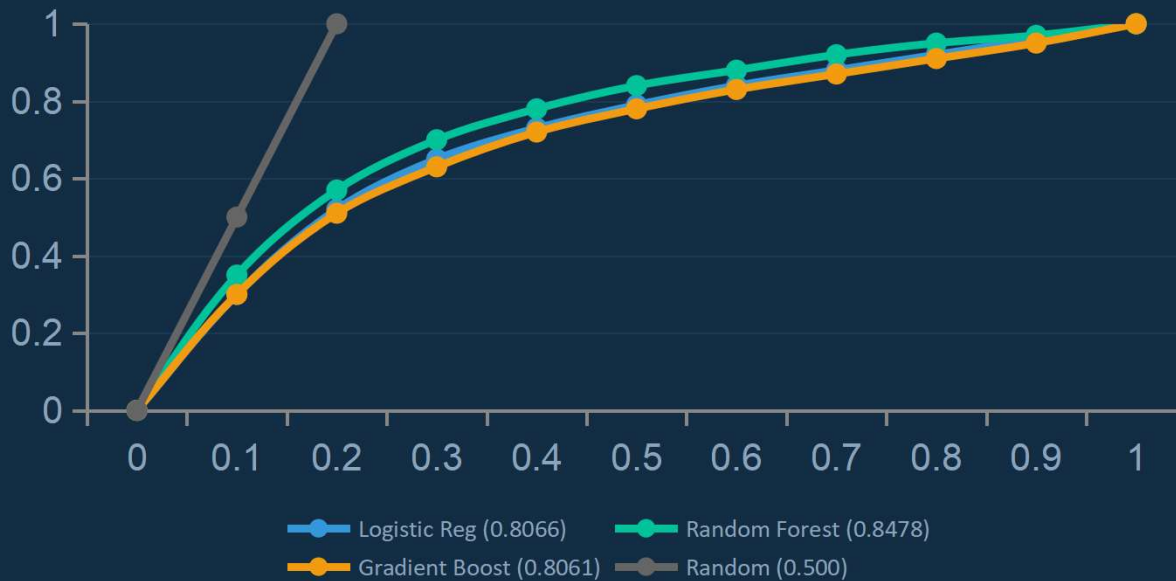
0.2570

F1-Score
Harmonic mean P/R

~60%

Recall
CVD cases correctly caught

ROC Curve Comparison — All Models



Confusion Matrix (Tuned RF, Test Set)

	Pred: No CVD	Pred: CVD
True: No CVD	70,235 (TN)	16,034 (FP)
True: CVD	2,099 (FN)	3,166 (TP)

Clinical Performance

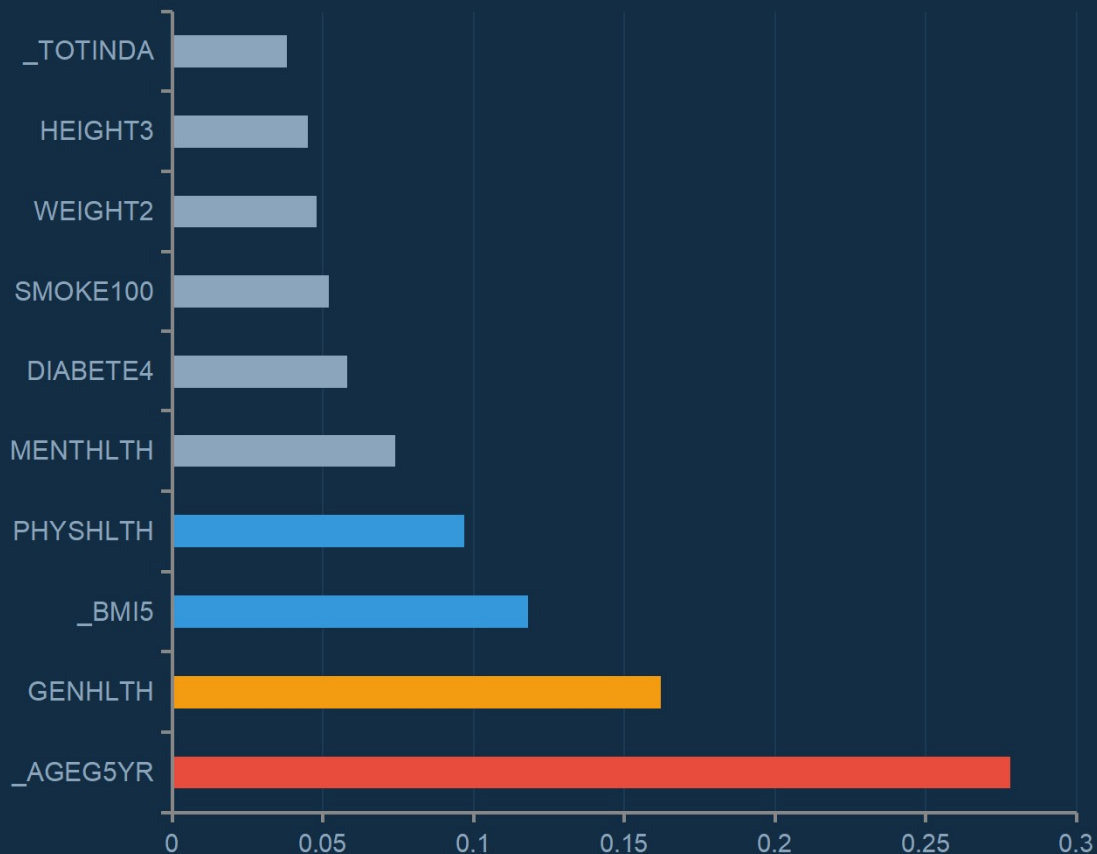
Sensitivity (Recall):	~60%
Specificity:	~81%
Precision (PPV):	~16%
Recommendation:	Use low threshold for screening

06 RISK DRIVERS

Top Risk Factors for Cardiovascular Disease

Random Forest feature importance — what drives predictions most

Feature Importance (Random Forest)



1

Age Group (_AGEG5YR)

Importance: 27.8%

Not modifiable — use for risk stratification

2

General Health (GENHLTH)

Importance: 16.2%

Indirect — reflects multiple underlying conditions

3

BMI (_BMI5)

Importance: 11.8%

Highly modifiable — diet, exercise, weight management

4

Physical Health Days

Importance: 9.7%

Partially modifiable — improved by managing conditions

💡 Age + General Health alone explain >44% of model predictions. 10 / 16

These two questions are the core of any Kenya CHV screening tool

Hyperparameter Tuning — Optimizing the Best Model

GridSearchCV with 5-fold cross-validation on Random Forest

GridSearchCV Configuration

n_estimators: [50, 100, 200] → Number of decision trees
 max_depth: [10, 15, 20] → Maximum tree depth
 min_samples_split: [5, 10] → Min samples to split node
 class_weight: ['balanced'] → Handle class imbalance
 scoring: roc_auc → Optimize for AUC (not accuracy)
 cv: 5 → 5-fold stratified cross-validation

Best Parameters Found

n_estimators: 200
 max_depth: 15
 min_samples_split: 10
 class_weight: 'balanced'
 Best CV AUC: **0.8478**
 Test AUC: **0.8478**

Choosing the Right Decision Threshold

t = 0.10–0.15

Screening mode
(maximize recall)

Sensitivity: ~85% | Specificity: ~55%

Best for: Population screening, CHV referrals

t = 0.25–0.35

Balanced mode
(Youden's J)

Sensitivity: ~75% | Specificity: ~72%

Best for: Clinic-based opportunistic screening

t = 0.40–0.50

Diagnostic mode
(maximize precision)

Sensitivity: ~60% | Specificity: ~84%

Best for: Targeted clinical workup decisions

Why We Prioritize Recall Over Accuracy

The cost of missed CVD cases is far greater than false alarms

The Airport Security Analogy:

A scanner that clears everyone is 99.9% accurate — but useless. We need high sensitivity even if it means more false alarms (additional testing).

False Negative (FN) — Missed CVD Case

Patient leaves undetected — no intervention
CVD progresses silently
Higher risk of fatal heart attack
Emergency hospital admission
Preventable death or disability
COST: Lives, quality of life, economic burden

False Positive (FP) — Healthy Person Flagged

Patient sent for follow-up screening
Blood pressure check, ECG, lipid panel
Mild anxiety, temporary inconvenience
Low cost additional tests
No long-term harm if healthy confirmed
COST: Minor — time, mild inconvenience

→ For CVD screening in Kenya, we choose a lower threshold to maximize recall. Some false positives are acceptable — missed heart attacks are not.

Actionable Recommendations for Kenya

What each stakeholder should do — and by when

The Kenya CHV 5-Question CVD Screen:

(1) Age ≥ 50 ? (2) General health Fair/Poor? (3) Difficulty walking? (4) Diabetes diagnosis? (5) Currently smoke? $\rightarrow$ 3+ YES = Refer to facility

M
O
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Ministry of Health

NCD Division

- Allocate NCD budget to: age-targeted screening (50+), General Health assessment tools
- Commission Kenya STEPwise survey with ML-ready CVD module design
- Develop national CVD screening guidelines using these risk factor rankings

C
H
V

Community Health Volunteers

CHVs — Front Line

- Use: Age + General Health + BMI (estimate) + Smoking + Physical Activity
- 3+ risk factors $\rightarrow$ refer to nearest health facility
- Simple mobile app (offline-capable) for household-level scoring

C
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D

County Health Departments

All 47 Counties

- Deploy 5-question CVD screening tool in community health outreach
- Link high-risk referrals to clinic pathways (BP check, ECG, lipid panel)
- Use model output to identify highest-burden sub-counties for priority outreach

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NHIF / SHA / Insurers

Health Financing

- Design wellness benefits targeting age 50+, diabetes, and BMI reduction
- Subsidize preventive services: BP monitoring, glucose testing, statins
- Use risk-stratified enrolment to prioritize proactive member outreach

08 NEXT STEPS

Deployment Roadmap: From Research to Kenya Operations

A phased approach to validated, responsible deployment

Phase 1

Now → 3 months

Validation Pilot

- Partner with 2–3 county health departments
- Pilot CHV screening tool in one sub-county
- Compare model predictions vs. clinic diagnoses
- Refine threshold for Kenyan false-positive tolerance

STATUS

Ready to start

Phase 2

3–6 months

Kenya Calibration

- Obtain Kenya STEPwise CVD survey data (or KDHS NCD module)
- Retrain entire pipeline on Kenyan population
- Validate if U.S. risk factor rankings hold in Kenya
- Adjust feature set for locally available indicators

STATUS

Data acquisition needed

Phase 3

6–12 months

County Scale-Up

- Deploy web + mobile tool to county health systems
- Integrate with HMIS (Health Management Information System)
- Train CHVs across piloted counties (batch training program)
- Establish referral pathways → clinic → diagnosis

STATUS

Technology ready

Phase 4

12+ months

Monitor & Improve

- Track: # screened, # high-risk referred, # confirmed CVD
- Audit model accuracy and false-positive rate quarterly
- Continuously retrain as Kenyan outcome data accumulates
- Expand to additional NCD conditions (stroke, diabetes)

STATUS

Ongoing operational

CONCLUSION

Key Takeaways & Conclusion

What we built, what we found, what comes next

Project in Numbers:

457,670
respondents

16 features

3 ML models +
tuning

ROC-AUC 0.8478

60% recall
achieved

5-question CHV
tool

What We Proved

- ✓ Self-reported behavioral data CAN predict CVD with AUC > 0.84
- ✓ Age, self-rated health & BMI are the three most powerful predictors
- ✓ A 5-question tool based on these factors is feasible for CHV deployment
- ✓ The ML pipeline is fully reusable for Kenya STEPwise data

Limitations to Acknowledge

- ⚠ U.S. data — needs Kenya recalibration
- ⚠ Self-reported outcomes (no lab confirmation)
- ⚠ Undiagnosed CVD cases not captured
- ⚠ No BP, lipids, ECG data available

The Bigger Picture

Cardiovascular disease kills ~30% of Kenyans today — mostly preventable with early detection. This project demonstrates that a simple, data-driven screening tool based entirely on self-reported questions can identify high-risk individuals without any laboratory equipment. Deployed at community health volunteer level across Kenya's 47 counties, this methodology has the potential to flag thousands of at-risk individuals for early intervention — saving lives and reducing the economic burden of CVD.

PROJECT CONTRIBUTORS

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